VAIMAANIK 2026
SUGAR ROCKET CHALLENGE

OFFICIAL COMPETITION RULEBOOK | Version 1.0

Global Academy of Technology, Bengaluru | 12-13 November 2026

1. Event Objective

The Sugar Rocket Challenge evaluates student rocket design through maximum horizontal distance, time of flight, and flight stability. Each team receives one official launch, subject to mandatory technical inspection, range approval, and applicable legal permissions.

2. Team Eligibility

1. Each team shall consist of exactly two (2) members.
2. Participants must carry valid college identification and complete event registration.
3. Each team must design and fabricate its own rocket airframe and present technical documentation at inspection.
4. Only one rocket and one official launch attempt are permitted per team.

3. Rocket and Motor Requirements

Item	Competition requirement
Rocket type	Student-designed model rocket airframe; one-stage configuration
Motor	Homemade sugar-propellant or certified commercial solid-propellant motor, subject to prior motor-specific and range safety approval
Homemade sugar motor	Permitted only with advance written permissions, specialist safety review, approved handling arrangements and explicit launch authorization; no on-site mixing or fabrication without separate authorization
Rocket dimensions and mass	Must fit the approved motor, launch system and the site safety limits announced by organizers
Guidance and launch system	Compatible launch rod/rail and approved remotely operated launch controller
Recovery	A safe recovery system is mandatory; recovery performance earns no points
Stages	Single stage only, unless organizers publish a separately authorized amendment

Both homemade sugar-propellant motors and commercially manufactured solid-propellant motors are eligible for consideration. Commercial motors require certification and event approval. Homemade motors require advance written authorization, documented safety review, proof of all applicable permissions for manufacture, storage, transport and use, and supervision by qualified experimental-rocketry personnel. No motor may be made, modified, loaded, tested or fired at the venue unless explicitly authorized under the approved safety plan.

4. Competition Format

1. One official launch per team. There are no elimination rounds and no best-of-two scoring.
2. Officials record the same launch for horizontal distance, elapsed flight time and stability.
3. All teams launch from the approved area under the direction of the range safety officer.
4. An organizer-confirmed pre-launch equipment malfunction may be reviewed before a valid launch is counted; a flight that has left the guide counts as the official attempt.
5. No relaunch shall be given for poor performance, weather changes, motor failure in flight or participant error.

5. Evaluation and Scoring

Criterion	Scoring
Horizontal distance	10 points for each complete 5 m
Flight time	10 points for each complete 1 s
Rocket flight stability	0-20 points awarded by judges
Total	Distance score + time score + stability score; no fixed overall maximum

5.1 Measurement Rules

- Distance is the horizontal distance from the launch point to the rocket's first ground-contact point, measured in metres.
- Flight time is measured from the rocket leaving the launcher until first ground contact, including the safe recovery descent, in seconds.
- Only fully completed 5-metre intervals and whole seconds earn points; partial intervals do not earn fractional points.
- Distance and flight time are recorded by designated officials using the same method for every team.

5.2 Calculation

Distance points = 10 × floor(distance in metres ÷ 5)

Time points = 10 × floor(flight time in seconds)

Final score = distance points + time points + stability points

“floor” means rounding down to the nearest whole number. Stability points cannot exceed 20.

5.3 Stability Rubric (20 points maximum)

Flight behaviour	Points
Straight and stable flight in the designated launch direction	16-20
Generally stable with minor deviations	11-15
Noticeable wobble or moderate deviation	6-10
Severe instability, tumbling or erratic flight	0-5

A normal ballistic arc is not by itself instability. The judges evaluate attitude control, wobble, tumbling and deviations from the approved launch direction.

6. Tie-Breaking

1. If total scores match, the greater actual horizontal distance ranks higher.
2. If distance matches, the greater actual flight duration ranks higher.
3. If both match, the higher stability score ranks higher.
4. If all remain equal, the teams share the position; no extra launch is required under the one-launch format.

7. Technical Inspection and Launch Procedure

1. Before launch, teams submit rocket details, motor type (homemade or commercial), supporting motor safety and authorization documents, and recovery-system information for inspection.
2. The qualified range safety officer checks structural integrity, stability documentation, motor eligibility and approvals, recovery provision, launch guide and ground equipment. Homemade motors require additional documented specialist review.
3. Only approved launch equipment and remotely operated ignition systems may be used. Organizers must inform teams in advance whether a compatible launch setup is supplied or must be submitted for approval.
4. Teams follow the announced queue and safety briefing. Access to the launch area is limited to authorized personnel.
5. The launch proceeds only after range clearance and an official go/no-go decision. Officials then record flight data and judge stability.
6. Recovery takes place only after the range is declared safe; spectators must not enter the launch or impact area.

8. Mandatory Safety Conditions

1. Commercially manufactured and homemade sugar-propellant motors may be entered only after required advance approvals and safety reviews. Unauthorized propellant manufacture, motor modification, pyrotechnic charges, unapproved ignition systems and any motor lacking required permissions are prohibited at the event.

2. No sharp or hazardous nose cones, unsecured heavy attachments or visibly damaged airframes are permitted.
3. Launches require institution approval, applicable local permissions, suitable range size, established exclusion zones and an appointed qualified range safety officer.
4. Fire protection, first aid, emergency communications and a stop-launch procedure must be available at the site.
5. Any unsafe wind, visibility, fire risk, malfunction or trajectory requires a hold or cancellation at the discretion of the range safety officer.
6. Any flight posing a risk to spectators, buildings, roads or other property may be declared invalid or disqualified.
7. There is no automatic right to launch: safety approval can be withdrawn at any time.

9. Disqualification

Disqualification may result from an unauthorized or unapproved motor or propellant; missing required documentation; an unsafe rocket or launcher; launching without clearance; violating exclusion zones; falsifying measurements or documents; interference with other teams; or failure to obey officials.

10. Results and Judges' Authority

Rankings are based on the final score from the single valid launch, after applying Section 6 tie-breakers. The judging panel verifies distance, time and stability points. Safety decisions by the range safety officer are final. The organizing committee may modify or cancel flights when required by safety, weather or authorization conditions.

11. Official Score Recording

Field	Entry
Team name / ID	_____
Team members	_____
Motor approval and inspection	Approved / Not approved
Horizontal distance (m)	_____
Flight time (s)	_____
Distance points	_____
Time points	_____
Stability points (0-20)	_____
Final total	_____
Judge / signature	_____

SAFETY NOTICE: Homemade motors may be launched only with documented legal permissions, qualified specialist approval and range safety clearance.